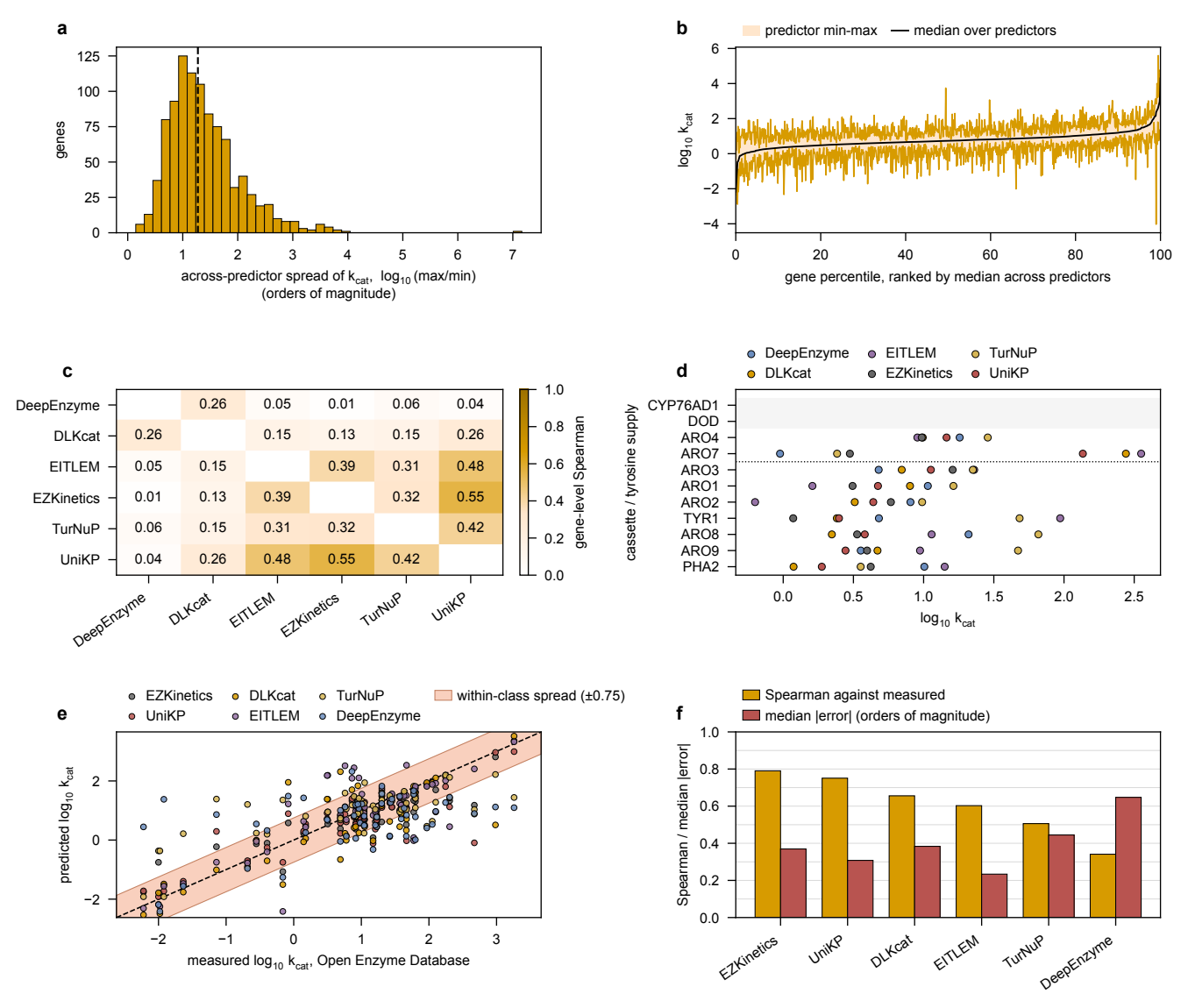

a
genes
across-predictor spread of k_{cat}, $\log_{10}$ (max/min)
(orders of magnitude)
b
predictor min-max
median over predictors
$\log_{10}$ k_{cat}
gene percentile, ranked by median across predictors
c
DeepEnzyme 0.26 0.05 0.01 0.06 0.04
DLKcat 0.26 0.15 0.13 0.15 0.26
EITLEM 0.05 0.15 0.39 0.31 0.48
EZKinetics 0.01 0.13 0.39 0.32 0.55
TurNuP 0.06 0.15 0.31 0.32 0.42
UniKP 0.04 0.26 0.48 0.55 0.42
gene-level Spearman
d
DeepEnzyme
DLKcat
EITLEM
EZKinetics
TurNuP
UniKP
CYP76AD1
DOD
ARO4
ARO7
ARO3
ARO1
ARO2
TYR1
ARO8
ARO9
PHA2
cassette / tyrosine supply
$\log_{10}$ k_{cat}
e
EZKinetics
DLKcat
TurNuP
UniKP
EITLEM
DeepEnzyme
within-class spread (±0.75)
predicted $\log_{10}$ k_{cat}
measured $\log_{10}$ k_{cat}, Open Enzyme Database
f
Spearman against measured
median |error| (orders of magnitude)
Spearman / median |error|
EZKinetics
UniKP
DLKcat
EITLEM
TurNuP
DeepEnzyme